Contents

- Comparing the Sample and Computational Efficiency of Unit-Precision Predictive Coding and Backpropagation
  Austin Huang **4**
- Benchmarking BRCA2 Missense Variant Pathogenicity Predictors Across Functional Domains and in a Subset of Variants Observed in Ashkenazi Jewish Individuals
  Meirav Bocknek, Mariame Diabate **16**
- Beyond Cycle Count: Sensor-Based Maintenance Warnings for Simulated Turbofan Engines
  Prahalad Talur **23**
- Dynamic Mathematical PDE Modeling and Machine Learning for TANGO2 Identification and Crisis Prediction
  Abhay Tippimath **29**
- Investigating the Efficacy of the Gemini 2.5 Pro Model in the Estimation of Macronutrient Content
  Sonali Guttikonda, Junyao Chen, He Tu, Zhengyuan Zhou **38**
- Analysis of Methods to Improve Underwater Object Detection in Various Environments
  Nico Katz **44**
- Optimizing Classification-Based Machine Learning Models for Bacterial Spot Disease Detection in Tomato Plants
  Siya Sharma, Emily Broadhurst **57**
- Demographic Sensitivity in Heartsteep Drink Recommendations: An Exploratory Audit of Three Language Models
  Yoomin (Rachel) Jung **62**
- Evaluating Representation Learning for Model-Independent Anomaly Detection in High-Energy Particle Collisions
  Abhijay Arun **72**
- Iterative Development of an AI Prototype for Dental Caries Detection and Localization with Informal Clinical Feedback
  Chinmay Karthik Amancha **88**
- Evaluating the Causal Faithfulness of SHAP as a Candidate Attribution Method for SAE Features
  Samuel Xie, Noura Al-Moubayed **94**